SAA Analysis LB data

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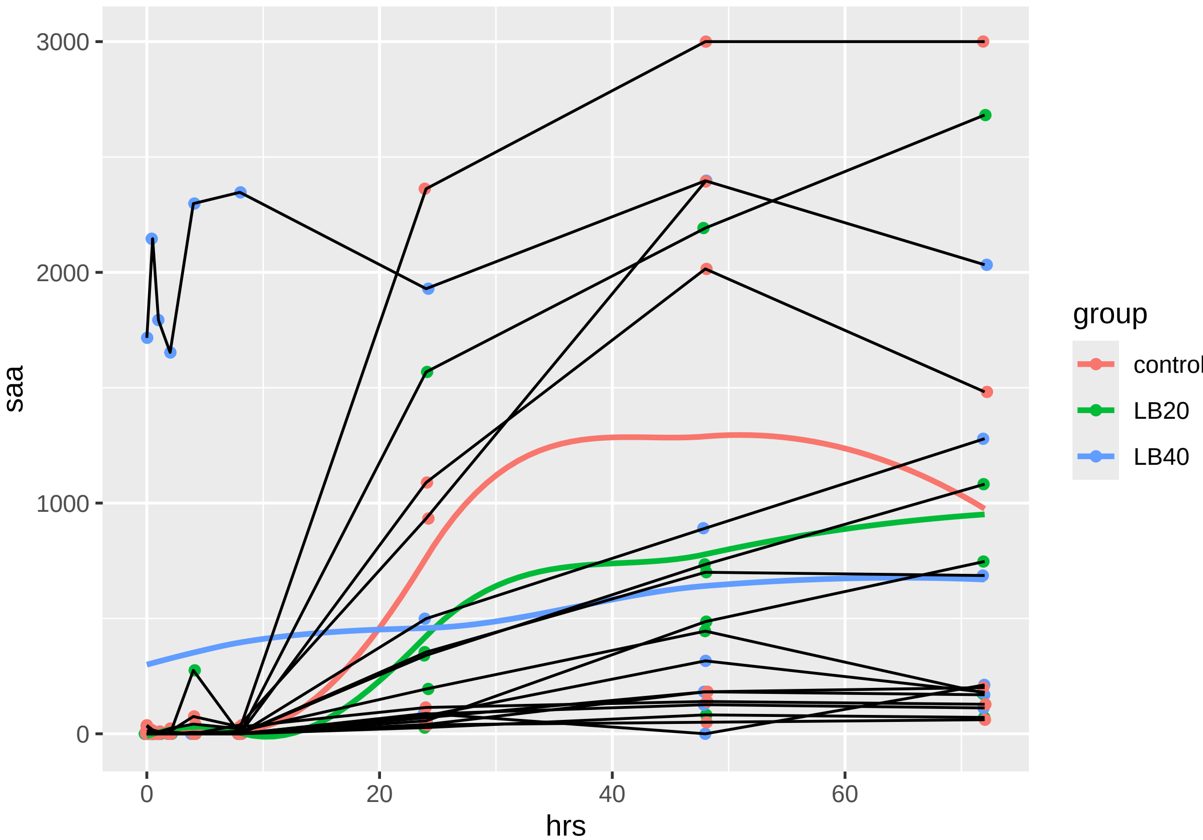
Import and check data

```
##    ID group hrs saa Eotaxin G.CSF IL1a IL.5 IL.18 IL.6 IL.17A IL.2 IL.4 IFNg
## 1 283  LB40 0.0   0      10    11   12   24  20.0   28     11   22   26   31
## 2 283  LB40 0.5   0      10    14   12   25  20.0   29     11   24   27   31
## 3 283  LB40 1.0   0      10    12   12   29  22.5   33     11   27   31   37
## 4 283  LB40 2.0   0      10    12   12   25  21.0   29     11   24   27   32
## 5 283  LB40 4.0   0      10    14   13   23  19.0   27     12   22   23   27
## 6 283  LB40 8.0   0       9    11   12   22  16.0   24     11   19   22   26
##   IL.8 IP.10 IL.10 TNFa
## 1   43    24    41   21
## 2  121    25    43   22
## 3  146    26    50   25
## 4  152    27    46   23
## 5  154    27    38   20
## 6   28    25    40   19
```

```
## 'data.frame':    161 obs. of  18 variables:
##  $ ID     : Factor w/ 18 levels "1079","1346",..: 3 3 3 3 3 3 3 3 3 17 ...
##  $ group  : Factor w/ 3 levels "control","LB20",..: 3 3 3 3 3 3 3 3 3 3 ...
##  $ hrs    : num  0 0.5 1 2 4 8 24 48 72 0 ...
##  $ saa    : int  0 0 0 0 0 0 70 182 169 0 ...
##  $ Eotaxin: num  10 10 10 10 10 9 9 10 9 9 ...
##  $ G.CSF  : num  11 14 12 12 14 11 15 13.5 11 11 ...
##  $ IL1a   : num  12 12 12 12 13 12 12 12 11 11 ...
##  $ IL.5   : num  24 25 29 25 23 22 25 22 22 14 ...
##  $ IL.18  : num  20 20 22.5 21 19 16 19 19 18 16 ...
##  $ IL.6   : num  28 29 33 29 27 24 26 27 25 17 ...
##  $ IL.17A : num  11 11 11 11 12 11 10 11 11 10 ...
##  $ IL.2   : num  22 24 27 24 22 19 22 21 20 13 ...
##  $ IL.4   : num  26 27 31 27 23 22 25 24 23 12 ...
##  $ IFNg   : num  31 31 37 32 27 26 30 27 27 17 ...
##  $ IL.8   : num  43 121 146 152 154 28 54 147 151 145 ...
##  $ IP.10  : num  24 25 26 27 27 25 21 27 24 20 ...
##  $ IL.10  : num  41 43 50 46 38 40 39 38 37 24 ...
##  $ TNFa   : num  21 22 25 23 20 19 21 19 18 12 ...
```

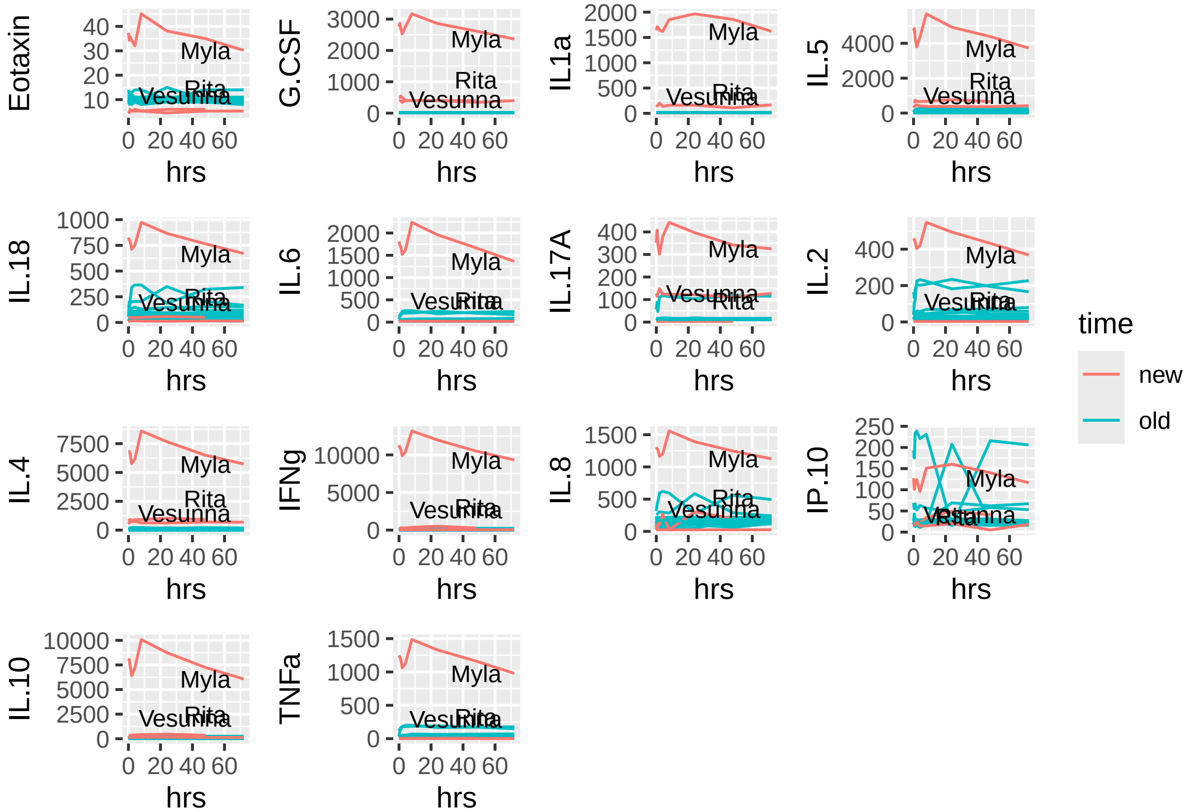
Plot all SAA data, with trend lines for each horse

```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



There is one outlier, check cytokine data for similar

```
## 'data.frame':    161 obs. of  19 variables:
##  $ ID     : Factor w/ 18 levels "1079","1346",..: 3 3 3 3 3 3 3 3 3 17 ...
##  $ group  : Factor w/ 3 levels "control","LB20",..: 3 3 3 3 3 3 3 3 3 3 ...
##  $ hrs    : num  0 0.5 1 2 4 8 24 48 72 0 ...
##  $ saa    : int  0 0 0 0 0 0 70 182 169 0 ...
##  $ Eotaxin: num  10 10 10 10 10 9 9 10 9 9 ...
##  $ G.CSF  : num  11 14 12 12 14 11 15 13.5 11 11 ...
##  $ IL1a   : num  12 12 12 12 13 12 12 12 11 11 ...
##  $ IL.5   : num  24 25 29 25 23 22 25 22 22 14 ...
##  $ IL.18  : num  20 20 22.5 21 19 16 19 19 18 16 ...
##  $ IL.6   : num  28 29 33 29 27 24 26 27 25 17 ...
##  $ IL.17A : num  11 11 11 11 12 11 10 11 11 10 ...
##  $ IL.2   : num  22 24 27 24 22 19 22 21 20 13 ...
##  $ IL.4   : num  26 27 31 27 23 22 25 24 23 12 ...
##  $ IFNg   : num  31 31 37 32 27 26 30 27 27 17 ...
##  $ IL.8   : num  43 121 146 152 154 28 54 147 151 145 ...
##  $ IP.10  : num  24 25 26 27 27 25 21 27 24 20 ...
##  $ IL.10  : num  41 43 50 46 38 40 39 38 37 24 ...
##  $ TNFa   : num  21 22 25 23 20 19 21 19 18 12 ...
##  $ time   : Factor w/ 2 levels "new","old": 2 2 2 2 2 2 2 2 2 2 ...
```



Similar, there seems to be a single outlier. Check formal test:

```
##
##  Dixon test for outliers
##
## data:  saa_0
## Q = 0.98019, p-value < 2.2e-16
## alternative hypothesis: highest value 1716 is an outlier

##
##  Dixon test for outliers
##
## data:  eotaxin_0
## Q = 0.86408, p-value < 2.2e-16
## alternative hypothesis: highest value 37.43 is an outlier

##
##  Dixon test for outliers
##
## data:  gcsf_0
## Q = 0.85789, p-value < 2.2e-16
## alternative hypothesis: highest value 2758 is an outlier

##
##  Dixon test for outliers
##
```

```
## data:  IL8_0
## Q = 0.8274, p-value < 2.2e-16
## alternative hypothesis: highest value 1307 is an outlier
```

Given the above analysis, suggest removing both horses from analysis. Both

were outliers at time 0, although interestingly, they were only outliers for one anaysis (either SAA or cytokines), but not both…

Remove outliers and summarize SAA data

```
## `summarise()` has grouped output by 'hrs'. You can override using the `.groups`
## argument.
```

hrs	group	mean	sd	min	max
0.0	control	9.00000	14.764823	0	34
	LB20	0.00000	0.000000	0	0
	LB40	0.00000	0.000000	0	0
0.5	control	4.20000	6.942622	0	16
	LB20	0.00000	0.000000	0	0
	LB40	0.00000	0.000000	0	0
1.0	control	2.00000	4.472136	0	10
	LB20	0.00000	0.000000	0	0
	LB40	0.00000	0.000000	0	0
2.0	control	3.80000	5.310367	0	11
	LB20	0.00000	0.000000	0	0
	LB40	0.00000	0.000000	0	0
4.0	control	15.00000	33.541020	0	75
	LB20	45.83333	112.268280	0	275
	LB40	1.40000	3.130495	0	7
8.0	control	13.60000	18.836136	0	38
	LB20	2.00000	4.898980	0	12
	LB40	0.00000	0.000000	0	0
24.0	control	443.00000	522.322219	39	1089
	LB20	423.16667	577.342331	27	1568
	LB40	164.20000	187.294421	70	499
48.0	control	956.20000	1148.378509	50	2394
	LB20	773.16667	733.190948	82	2192
	LB40	303.00000	347.739845	0	891
72.0	control	467.75000	678.544705	61	1482
	LB20	951.60000	1052.397406	71	2682
	LB40	439.66667	461.529053	112	1279

Create model

```
##                  Estimate Std. Error        df    t value     Pr(>|t|)
## (Intercept)     16.451133 103.842162  17.09862  0.1584244 8.759784e-01
```

```
## groupLB20     -40.467028 140.221415  17.84907 -0.2885938 7.762170e-01
## groupLB40     -16.113874 146.749954  17.90242 -0.1098050 9.137856e-01
## hrs            11.821912   2.241070 124.71596  5.2751199 5.674401e-07
## groupLB20:hrs   2.817595   3.018326 124.62129  0.9334960 3.523691e-01
## groupLB40:hrs  -5.490268   3.010588 125.64803 -1.8236532 7.058122e-02
```

The only differences are in time, no groups are different overall

Check if there are differences between groups at any time point?

```
## hrs =  0.0:
##  contrast       estimate  SE   df t.ratio p.value
##  control - LB20    40.47 140 17.9   0.289  0.9552
##  control - LB40    16.11 147 17.9   0.110  0.9934
##  LB20 - LB40      -24.35 140 21.0  -0.174  0.9834
##
## hrs =  0.5:
##  contrast       estimate  SE   df t.ratio p.value
##  control - LB20    39.06 140 17.6   0.280  0.9579
##  control - LB40    18.86 146 17.7   0.129  0.9909
##  LB20 - LB40      -20.20 139 20.7  -0.145  0.9885
##
## hrs =  1.0:
##  contrast       estimate  SE   df t.ratio p.value
##  control - LB20    37.65 139 17.3   0.271  0.9606
##  control - LB40    21.60 146 17.4   0.148  0.9879
##  LB20 - LB40      -16.05 139 20.4  -0.116  0.9926
##
## hrs =  2.0:
##  contrast       estimate  SE   df t.ratio p.value
##  control - LB20    34.83 138 16.9   0.252  0.9657
##  control - LB40    27.09 145 17.0   0.187  0.9808
##  LB20 - LB40       -7.74 137 19.9  -0.056  0.9983
##
## hrs =  4.0:
##  contrast       estimate  SE   df t.ratio p.value
##  control - LB20    29.20 136 16.0   0.214  0.9751
##  control - LB40    38.07 143 16.2   0.267  0.9615
##  LB20 - LB40        8.88 135 19.0   0.066  0.9976
##
## hrs =  8.0:
##  contrast       estimate  SE   df t.ratio p.value
##  control - LB20    17.93 134 14.7   0.134  0.9901
##  control - LB40    60.04 139 14.9   0.431  0.9034
##  LB20 - LB40       42.11 132 17.6   0.319  0.9457
##
## hrs = 24.0:
##  contrast       estimate  SE   df t.ratio p.value
##  control - LB20   -27.16 133 14.4  -0.204  0.9774
##  control - LB40   147.88 137 14.4   1.083  0.5391
##  LB20 - LB40      175.04 128 17.1   1.366  0.3799
##
## hrs = 48.0:
```

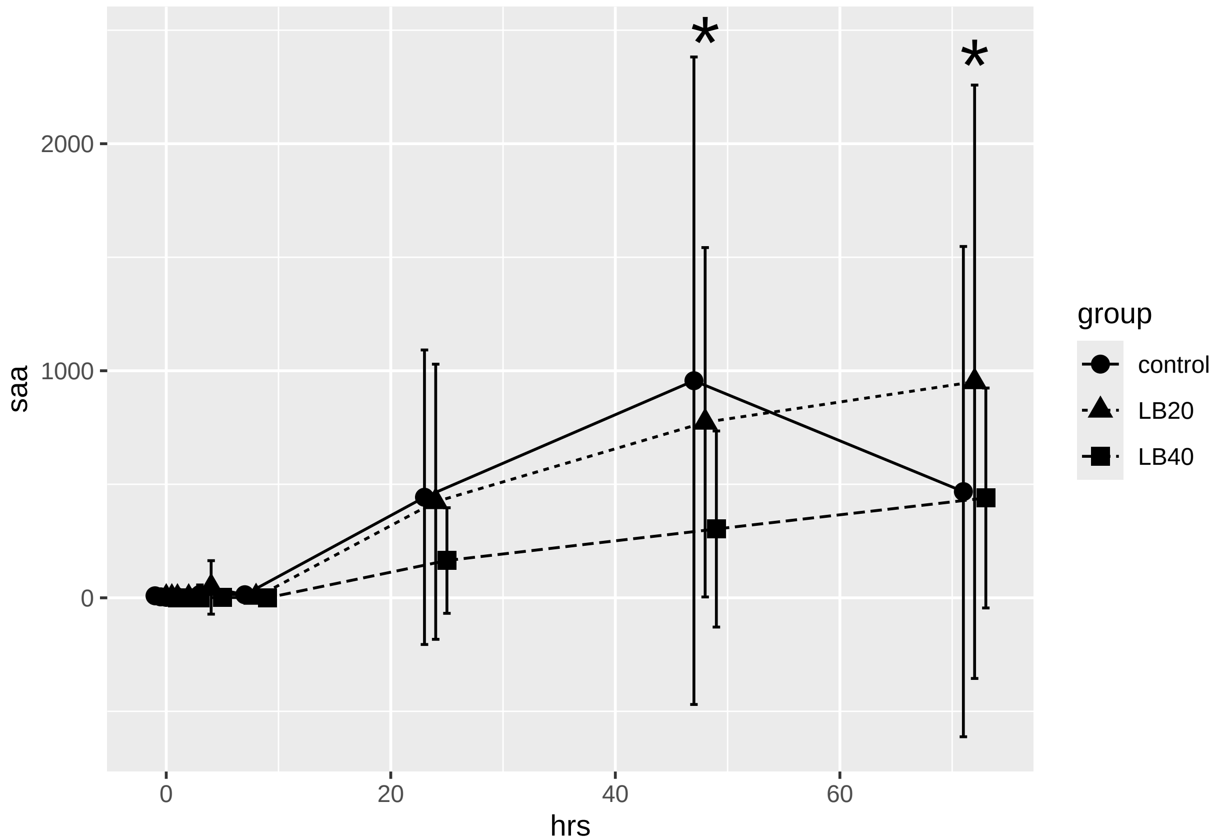
```
##  contrast       estimate  SE   df t.ratio p.value
##  control - LB20   -94.78 162 30.5  -0.585  0.8293
##  control - LB40   279.65 162 29.6   1.728  0.2118
##  LB20 - LB40      374.42 151 36.7   2.482  0.0457
##
## hrs = 72.0:
##  contrast       estimate  SE   df t.ratio p.value
##  control - LB20  -162.40 213 71.8  -0.763  0.7270
##  control - LB40   411.41 210 71.4   1.957  0.1304
##  LB20 - LB40      573.81 196 87.3   2.924  0.0121
##
## Degrees-of-freedom method: kenward-roger
## P value adjustment: tukey method for comparing a family of 3 estimates
```

It looks like the control and LB40 groups are different at 48 and 72 hrs (with

40 being lower)

Plot the differences

```
## ------------------------------------------------------------------------------

## You have loaded plyr after dplyr - this is likely to cause problems.
## If you need functions from both plyr and dplyr, please load plyr first, then dplyr:
## library(plyr); library(dplyr)

## ------------------------------------------------------------------------------

##
## Attaching package: 'plyr'

## The following objects are masked from 'package:dplyr':
##
##     arrange, count, desc, failwith, id, mutate, rename, summarise,
##     summarize

## The following object is masked from 'package:purrr':
##
##     compact
```



There were small differences between

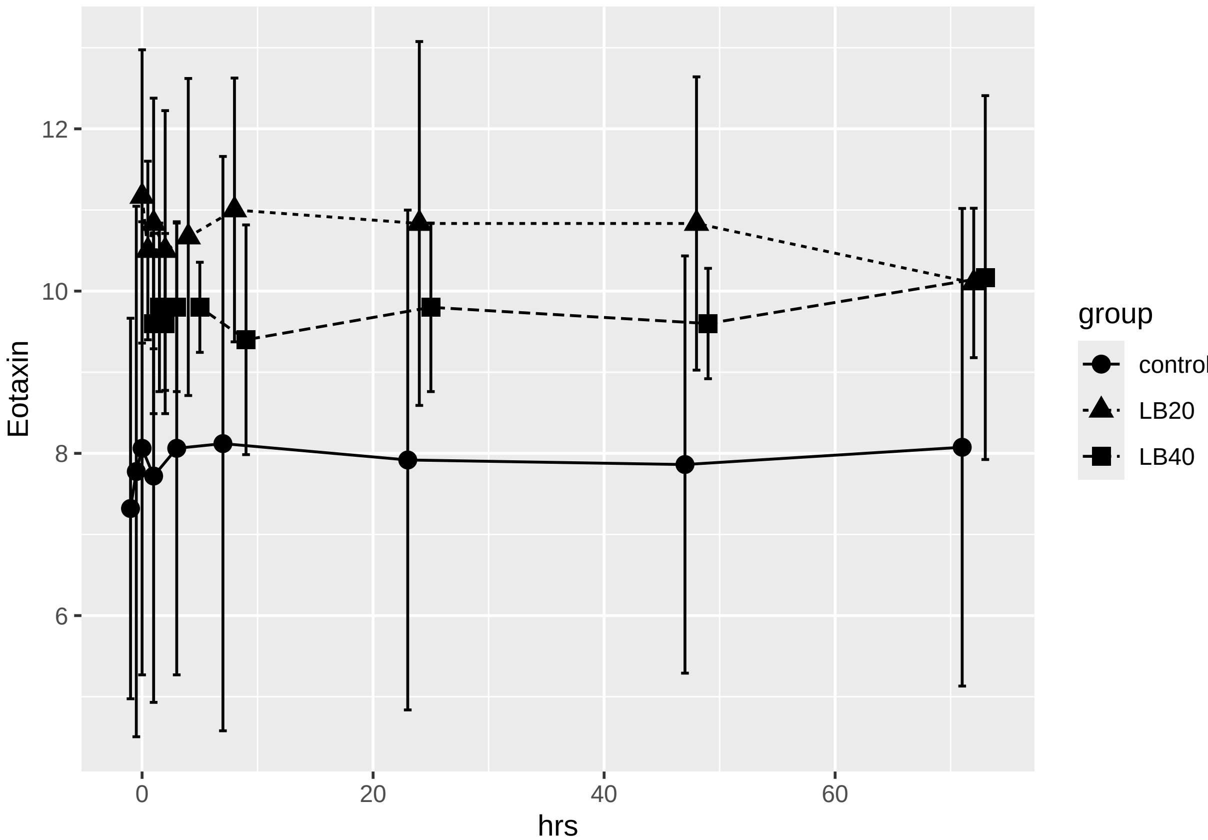
groups for Eotaxin, G-CSF and IL-8

Eotaxin - all time points control was different from both groups except at time 72 hrs, where LB20 was not different from control

```
##                    Estimate  Std. Error        df     t value     Pr(>|t|)
## (Intercept)    7.8598420028 0.726586462  13.69412 10.81749029 4.344792e-08
## groupLB20      2.3457239975 0.932759631  15.56378  2.51482152 2.332212e-02
## groupLB40      2.5369175514 0.959737100  16.44970  2.64334634 1.741274e-02
## hrs           -0.0021167459 0.005172880 123.28287 -0.40920064 6.831025e-01
## groupLB20:hrs  0.0005167564 0.006966649 123.29825  0.07417575 9.409907e-01
## groupLB40:hrs  0.0007022010 0.007024196 125.87617  0.09996887 9.205281e-01

## hrs =  0.0:
##  contrast       estimate    SE   df t.ratio p.value
##  control - LB20   -2.346 0.933 15.6  -2.515  0.0574
##  control - LB40   -2.537 0.960 16.4  -2.643  0.0436
##  LB20 - LB40      -0.191 0.711 61.2  -0.269  0.9610
##
## hrs =  0.5:
##  contrast       estimate    SE   df t.ratio p.value
##  control - LB20   -2.346 0.932 15.5  -2.516  0.0573
##  control - LB40   -2.537 0.959 16.4  -2.646  0.0435
##  LB20 - LB40      -0.191 0.710 61.1  -0.269  0.9608
```

```
##
## hrs =  1.0:
##  contrast       estimate    SE   df t.ratio p.value
##  control - LB20   -2.346 0.932 15.5  -2.518  0.0572
##  control - LB40   -2.538 0.958 16.4  -2.648  0.0434
##  LB20 - LB40      -0.191 0.709 60.9  -0.270  0.9607
##
## hrs =  2.0:
##  contrast       estimate    SE   df t.ratio p.value
##  control - LB20   -2.347 0.931 15.5  -2.520  0.0570
##  control - LB40   -2.538 0.957 16.3  -2.652  0.0431
##  LB20 - LB40      -0.192 0.707 60.7  -0.271  0.9604
##
## hrs =  4.0:
##  contrast       estimate    SE   df t.ratio p.value
##  control - LB20   -2.348 0.930 15.4  -2.525  0.0567
##  control - LB40   -2.540 0.955 16.2  -2.660  0.0426
##  LB20 - LB40      -0.192 0.703 60.2  -0.273  0.9598
##
## hrs =  8.0:
##  contrast       estimate    SE   df t.ratio p.value
##  control - LB20   -2.350 0.928 15.2  -2.533  0.0561
##  control - LB40   -2.543 0.951 16.0  -2.674  0.0417
##  LB20 - LB40      -0.193 0.697 59.5  -0.277  0.9587
##
## hrs = 24.0:
##  contrast       estimate    SE   df t.ratio p.value
##  control - LB20   -2.358 0.928 15.2  -2.541  0.0552
##  control - LB40   -2.554 0.943 15.7  -2.708  0.0394
##  LB20 - LB40      -0.196 0.679 59.3  -0.288  0.9553
##
## hrs = 48.0:
##  contrast       estimate    SE   df t.ratio p.value
##  control - LB20   -2.371 0.953 16.9  -2.487  0.0583
##  control - LB40   -2.571 0.956 16.9  -2.689  0.0392
##  LB20 - LB40      -0.200 0.684 67.9  -0.292  0.9540
##
## hrs = 72.0:
##  contrast       estimate    SE   df t.ratio p.value
##  control - LB20   -2.383 1.010 20.8  -2.370  0.0680
##  control - LB40   -2.587 0.998 20.5  -2.593  0.0435
##  LB20 - LB40      -0.205 0.726 87.0  -0.282  0.9572
##
## Degrees-of-freedom method: kenward-roger
## P value adjustment: tukey method for comparing a family of 3 estimates

## hrs =  0.0:
##  contrast       estimate    SE   df t.ratio p.value
##  LB20 - control     2.35 0.933 15.6   2.515  0.0438
##  LB40 - control     2.54 0.960 16.4   2.643  0.0329
##
## hrs =  0.5:
##  contrast       estimate    SE   df t.ratio p.value
##  LB20 - control     2.35 0.932 15.5   2.516  0.0438
```

```
##  LB40 - control     2.54 0.959 16.4   2.646  0.0328
##
## hrs =  1.0:
##  contrast       estimate    SE   df t.ratio p.value
##  LB20 - control     2.35 0.932 15.5   2.518  0.0437
##  LB40 - control     2.54 0.958 16.4   2.648  0.0327
##
## hrs =  2.0:
##  contrast       estimate    SE   df t.ratio p.value
##  LB20 - control     2.35 0.931 15.5   2.520  0.0435
##  LB40 - control     2.54 0.957 16.3   2.652  0.0326
##
## hrs =  4.0:
##  contrast       estimate    SE   df t.ratio p.value
##  LB20 - control     2.35 0.930 15.4   2.525  0.0433
##  LB40 - control     2.54 0.955 16.2   2.660  0.0322
##
## hrs =  8.0:
##  contrast       estimate    SE   df t.ratio p.value
##  LB20 - control     2.35 0.928 15.2   2.533  0.0428
##  LB40 - control     2.54 0.951 16.0   2.674  0.0315
##
## hrs = 24.0:
##  contrast       estimate    SE   df t.ratio p.value
##  LB20 - control     2.36 0.928 15.2   2.541  0.0422
##  LB40 - control     2.55 0.943 15.7   2.708  0.0297
##
## hrs = 48.0:
##  contrast       estimate    SE   df t.ratio p.value
##  LB20 - control     2.37 0.953 16.9   2.487  0.0445
##  LB40 - control     2.57 0.956 16.9   2.689  0.0295
##
## hrs = 72.0:
##  contrast       estimate    SE   df t.ratio p.value
##  LB20 - control     2.38 1.010 20.8   2.370  0.0518
##  LB40 - control     2.59 0.998 20.5   2.593  0.0326
##
## Degrees-of-freedom method: kenward-roger
## P value adjustment: dunnettx method for 2 tests
```



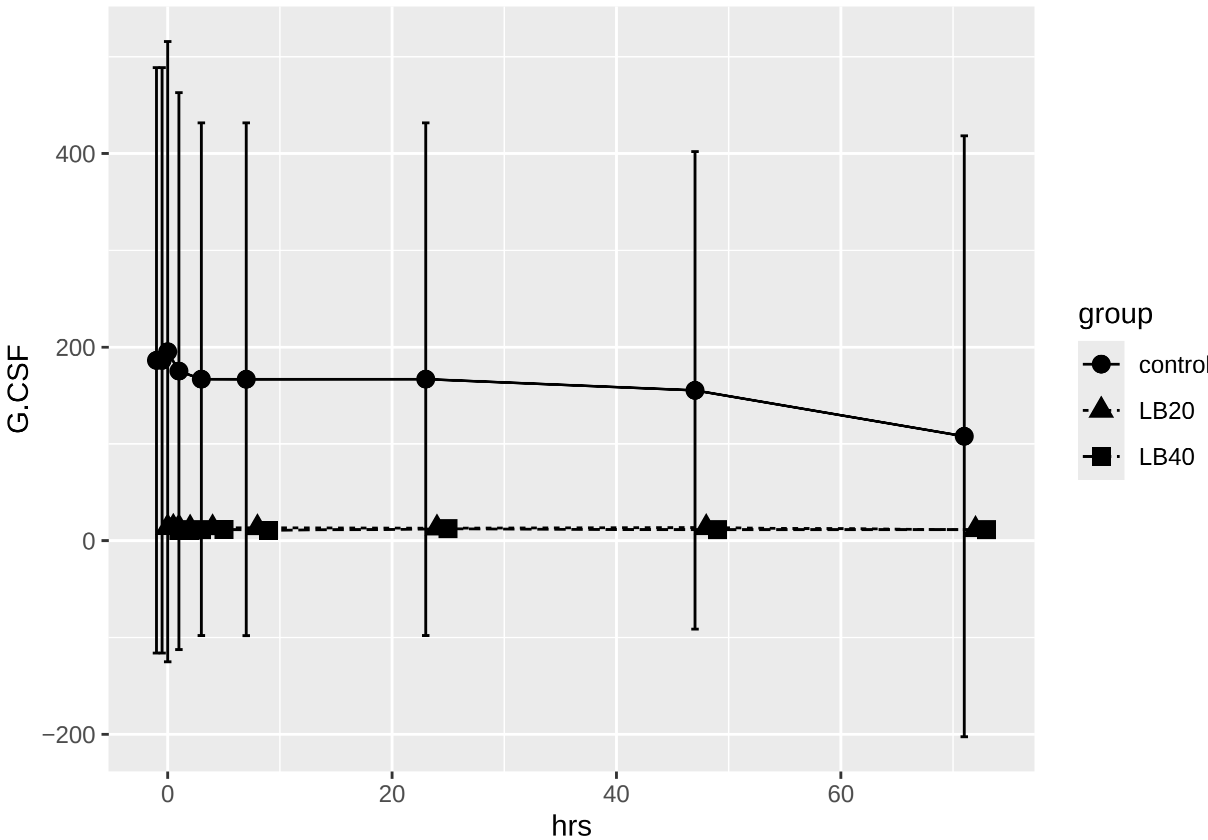
G-CSF

```
##                    Estimate Std. Error        df   t value    Pr(>|t|)
## (Intercept)    179.9035173 53.5425835  13.90980  3.360008 0.004707258
## groupLB20     -167.0696947 65.1675658  14.38453 -2.563694 0.022145654
## groupLB40     -167.9168022 65.4634222  14.62511 -2.565048 0.021869252
## hrs             -0.3746326  0.1137836 123.01212 -3.292503 0.001296579
## groupLB20:hrs    0.3639187  0.1532492 123.01772  2.374686 0.019109988
## groupLB40:hrs    0.3699176  0.1556688 123.64752  2.376312 0.019021786

## hrs =  0.0:
##  contrast       estimate   SE    df t.ratio p.value
##  control - LB20  167.070 65.2  14.4   2.564  0.0544
##  control - LB40  167.917 65.5  14.6   2.565  0.0538
##  LB20 - LB40       0.847 19.5 134.8   0.043  0.9990
##
## hrs =  0.5:
##  contrast       estimate   SE    df t.ratio p.value
##  control - LB20  166.888 65.2  14.4   2.561  0.0547
##  control - LB40  167.732 65.5  14.6   2.562  0.0541
##  LB20 - LB40       0.844 19.5 134.8   0.043  0.9990
##
## hrs =  1.0:
##  contrast       estimate   SE    df t.ratio p.value
##  control - LB20  166.706 65.2  14.4   2.558  0.0550
##  control - LB40  167.547 65.5  14.6   2.560  0.0544
##  LB20 - LB40       0.841 19.5 134.8   0.043  0.9990
```

```
##
## hrs =  2.0:
##  contrast       estimate   SE    df t.ratio p.value
##  control - LB20  166.342 65.2  14.4   2.553  0.0555
##  control - LB40  167.177 65.4  14.6   2.555  0.0549
##  LB20 - LB40       0.835 19.4 134.8   0.043  0.9990
##
## hrs =  4.0:
##  contrast       estimate   SE    df t.ratio p.value
##  control - LB20  165.614 65.1  14.4   2.542  0.0567
##  control - LB40  166.437 65.4  14.6   2.544  0.0561
##  LB20 - LB40       0.823 19.3 134.9   0.043  0.9990
##
## hrs =  8.0:
##  contrast       estimate   SE    df t.ratio p.value
##  control - LB20  164.158 65.1  14.4   2.520  0.0590
##  control - LB40  164.957 65.4  14.6   2.523  0.0584
##  LB20 - LB40       0.799 19.2 134.9   0.042  0.9990
##
## hrs = 24.0:
##  contrast       estimate   SE    df t.ratio p.value
##  control - LB20  158.336 65.1  14.4   2.431  0.0696
##  control - LB40  159.039 65.3  14.5   2.435  0.0687
##  LB20 - LB40       0.703 18.7 134.9   0.038  0.9992
##
## hrs = 48.0:
##  contrast       estimate   SE    df t.ratio p.value
##  control - LB20  149.602 65.3  14.5   2.290  0.0892
##  control - LB40  150.161 65.4  14.5   2.298  0.0880
##  LB20 - LB40       0.559 18.6 134.5   0.030  0.9995
##
## hrs = 72.0:
##  contrast       estimate   SE    df t.ratio p.value
##  control - LB20  140.868 65.7  14.9   2.144  0.1146
##  control - LB40  141.283 65.6  14.8   2.153  0.1130
##  LB20 - LB40       0.415 19.1 133.5   0.022  0.9997
##
## Degrees-of-freedom method: kenward-roger
## P value adjustment: tukey method for comparing a family of 3 estimates
```

```
## hrs =  0.0:
##  contrast       estimate   SE   df t.ratio p.value
##  LB20 - control     -167 65.2 14.4  -2.564  0.0416
##  LB40 - control     -168 65.5 14.6  -2.565  0.0411
##
## hrs =  0.5:
##  contrast       estimate   SE   df t.ratio p.value
##  LB20 - control     -167 65.2 14.4  -2.561  0.0418
##  LB40 - control     -168 65.5 14.6  -2.562  0.0413
##
## hrs =  1.0:
##  contrast       estimate   SE   df t.ratio p.value
##  LB20 - control     -167 65.2 14.4  -2.558  0.0421
##  LB40 - control     -168 65.5 14.6  -2.560  0.0416
```

```
##
## hrs =  2.0:
##  contrast       estimate   SE   df t.ratio p.value
##  LB20 - control     -166 65.2 14.4  -2.553  0.0425
##  LB40 - control     -167 65.4 14.6  -2.555  0.0420
##
## hrs =  4.0:
##  contrast       estimate   SE   df t.ratio p.value
##  LB20 - control     -166 65.1 14.4  -2.542  0.0434
##  LB40 - control     -166 65.4 14.6  -2.544  0.0429
##
## hrs =  8.0:
##  contrast       estimate   SE   df t.ratio p.value
##  LB20 - control     -164 65.1 14.4  -2.520  0.0453
##  LB40 - control     -165 65.4 14.6  -2.523  0.0447
##
## hrs = 24.0:
##  contrast       estimate   SE   df t.ratio p.value
##  LB20 - control     -158 65.1 14.4  -2.431  0.0537
##  LB40 - control     -159 65.3 14.5  -2.435  0.0530
##
## hrs = 48.0:
##  contrast       estimate   SE   df t.ratio p.value
##  LB20 - control     -150 65.3 14.5  -2.290  0.0695
##  LB40 - control     -150 65.4 14.5  -2.298  0.0685
##
## hrs = 72.0:
##  contrast       estimate   SE   df t.ratio p.value
##  LB20 - control     -141 65.7 14.9  -2.144  0.0902
##  LB40 - control     -141 65.6 14.8  -2.153  0.0889
##
## Degrees-of-freedom method: kenward-roger
## P value adjustment: dunnettx method for 2 tests
```



IL-8

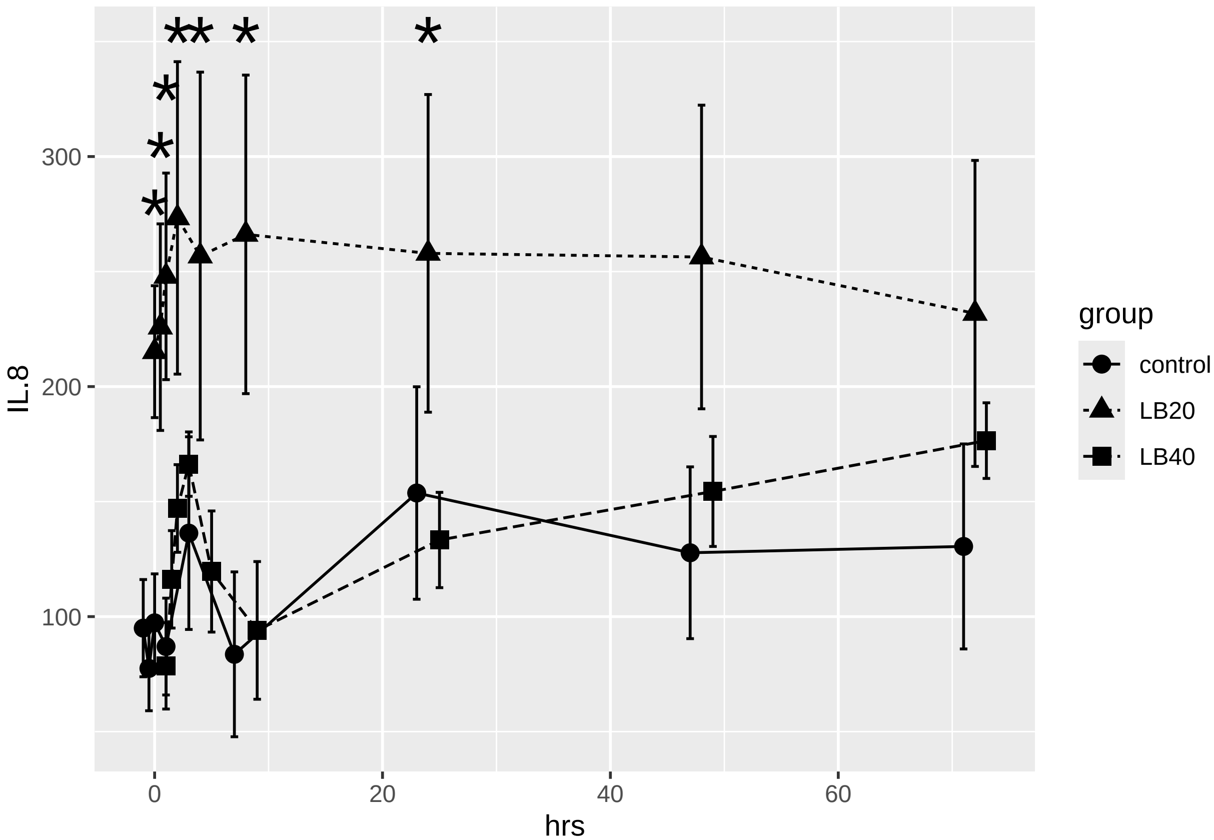
```
##                     Estimate Std. Error         df    t value   Pr(>|t|)
## (Intercept)    98.31559065 39.3073871  13.87365  2.5011988 0.02553973
## groupLB20     148.23576275 51.4924838  15.65638  2.8787845 0.01109609
## groupLB40      22.27167551 53.4070875  16.42545  0.4170172 0.68206513
## hrs             0.72245445  0.3813399 123.52717  1.8945159 0.06049441
## groupLB20:hrs  -0.62237658  0.5135627 123.53147 -1.2118804 0.22787110
## groupLB40:hrs  -0.07990687  0.5159031 126.06673 -0.1548874 0.87715783
```

```
## hrs =  0.0:
##  contrast       estimate   SE   df t.ratio p.value
##  control - LB20   -148.2 51.5 15.7  -2.879  0.0283
##  control - LB40    -22.3 53.4 16.4  -0.417  0.9091
##  LB20 - LB40       126.0 44.7 37.8   2.817  0.0205
##
## hrs =  0.5:
##  contrast       estimate   SE   df t.ratio p.value
##  control - LB20   -147.9 51.5 15.6  -2.875  0.0286
##  control - LB40    -22.2 53.3 16.4  -0.417  0.9092
##  LB20 - LB40       125.7 44.6 37.6   2.816  0.0206
##
## hrs =  1.0:
##  contrast       estimate   SE   df t.ratio p.value
##  control - LB20   -147.6 51.4 15.6  -2.871  0.0288
##  control - LB40    -22.2 53.3 16.3  -0.416  0.9094
```

```
##  LB20 - LB40       125.4 44.6 37.5   2.814  0.0207
##
## hrs =  2.0:
##  contrast       estimate   SE   df t.ratio p.value
##  control - LB20   -147.0 51.3 15.5  -2.863  0.0294
##  control - LB40    -22.1 53.2 16.2  -0.416  0.9096
##  LB20 - LB40       124.9 44.4 37.3   2.811  0.0209
##
## hrs =  4.0:
##  contrast       estimate   SE   df t.ratio p.value
##  control - LB20   -145.7 51.2 15.3  -2.846  0.0306
##  control - LB40    -22.0 53.0 16.0  -0.414  0.9102
##  LB20 - LB40       123.8 44.1 36.8   2.804  0.0213
##
## hrs =  8.0:
##  contrast       estimate   SE   df t.ratio p.value
##  control - LB20   -143.3 51.0 15.0  -2.809  0.0333
##  control - LB40    -21.6 52.6 15.7  -0.411  0.9116
##  LB20 - LB40       121.6 43.7 36.1   2.786  0.0225
##
## hrs = 24.0:
##  contrast       estimate   SE   df t.ratio p.value
##  control - LB20   -133.3 51.0 15.0  -2.614  0.0485
##  control - LB40    -20.4 52.1 15.4  -0.391  0.9196
##  LB20 - LB40       112.9 42.5 35.9   2.655  0.0309
##
## hrs = 48.0:
##  contrast       estimate   SE   df t.ratio p.value
##  control - LB20   -118.4 53.4 18.0  -2.216  0.0955
##  control - LB40    -18.4 53.6 18.0  -0.344  0.9371
##  LB20 - LB40        99.9 43.6 44.9   2.293  0.0670
##
## hrs = 72.0:
##  contrast       estimate   SE   df t.ratio p.value
##  control - LB20   -103.4 58.4 25.4  -1.771  0.1995
##  control - LB40    -16.5 57.8 25.0  -0.286  0.9561
##  LB20 - LB40        86.9 47.6 67.2   1.827  0.1687
##
## Degrees-of-freedom method: kenward-roger
## P value adjustment: tukey method for comparing a family of 3 estimates
```

```
## hrs =  0.0:
##  contrast       estimate   SE   df t.ratio p.value
##  LB20 - control    148.2 51.5 15.7   2.879  0.0211
##  LB40 - control     22.3 53.4 16.4   0.417  0.8703
##
## hrs =  0.5:
##  contrast       estimate   SE   df t.ratio p.value
##  LB20 - control    147.9 51.5 15.6   2.875  0.0213
##  LB40 - control     22.2 53.3 16.4   0.417  0.8705
##
## hrs =  1.0:
##  contrast       estimate   SE   df t.ratio p.value
##  LB20 - control    147.6 51.4 15.6   2.871  0.0216
```

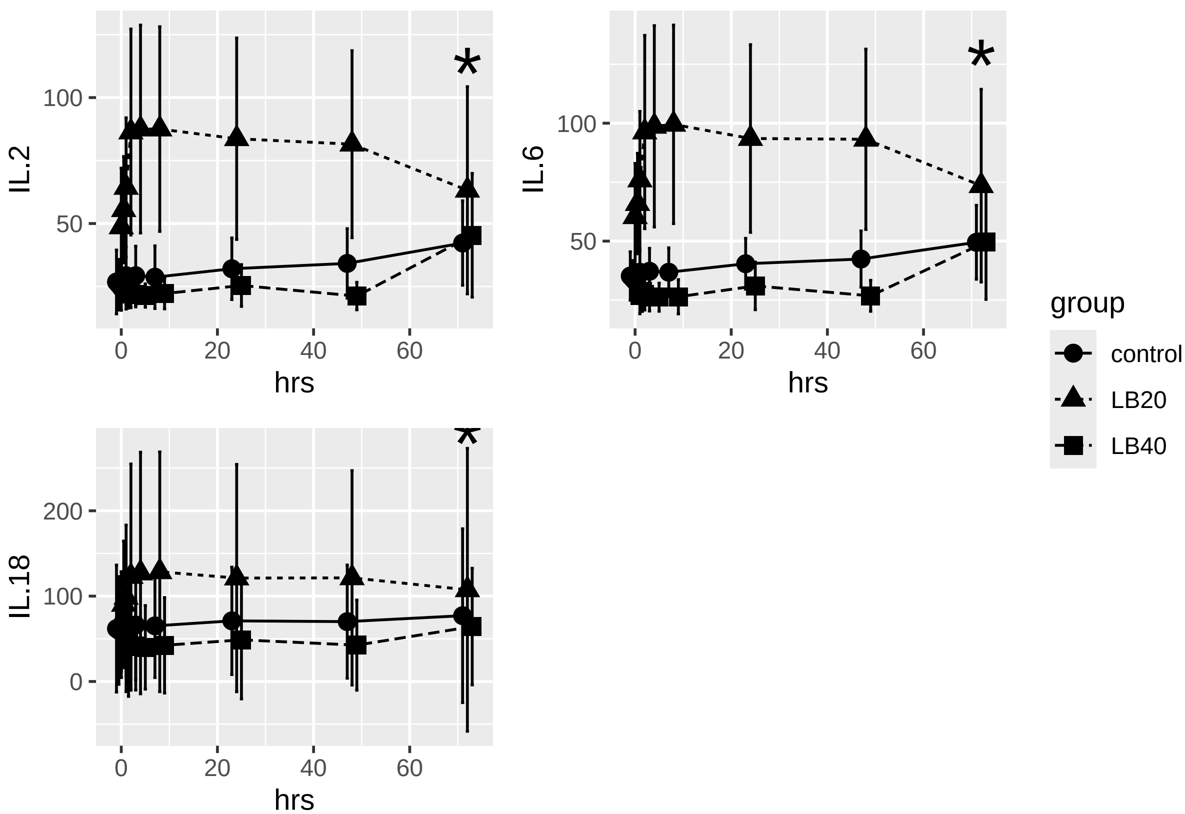
```
##  LB40 - control     22.2 53.3 16.3   0.416  0.8706
##
## hrs =  2.0:
##  contrast       estimate   SE   df t.ratio p.value
##  LB20 - control    147.0 51.3 15.5   2.863  0.0220
##  LB40 - control     22.1 53.2 16.2   0.416  0.8710
##
## hrs =  4.0:
##  contrast       estimate   SE   df t.ratio p.value
##  LB20 - control    145.7 51.2 15.3   2.846  0.0230
##  LB40 - control     22.0 53.0 16.0   0.414  0.8717
##
## hrs =  8.0:
##  contrast       estimate   SE   df t.ratio p.value
##  LB20 - control    143.3 51.0 15.0   2.809  0.0250
##  LB40 - control     21.6 52.6 15.7   0.411  0.8735
##
## hrs = 24.0:
##  contrast       estimate   SE   df t.ratio p.value
##  LB20 - control    133.3 51.0 15.0   2.614  0.0369
##  LB40 - control     20.4 52.1 15.4   0.391  0.8834
##
## hrs = 48.0:
##  contrast       estimate   SE   df t.ratio p.value
##  LB20 - control    118.4 53.4 18.0   2.216  0.0740
##  LB40 - control     18.4 53.6 18.0   0.344  0.9054
##
## hrs = 72.0:
##  contrast       estimate   SE   df t.ratio p.value
##  LB20 - control    103.4 58.4 25.4   1.771  0.1597
##  LB40 - control     16.5 57.8 25.0   0.286  0.9302
##
## Degrees-of-freedom method: kenward-roger
## P value adjustment: dunnettx method for 2 tests
```



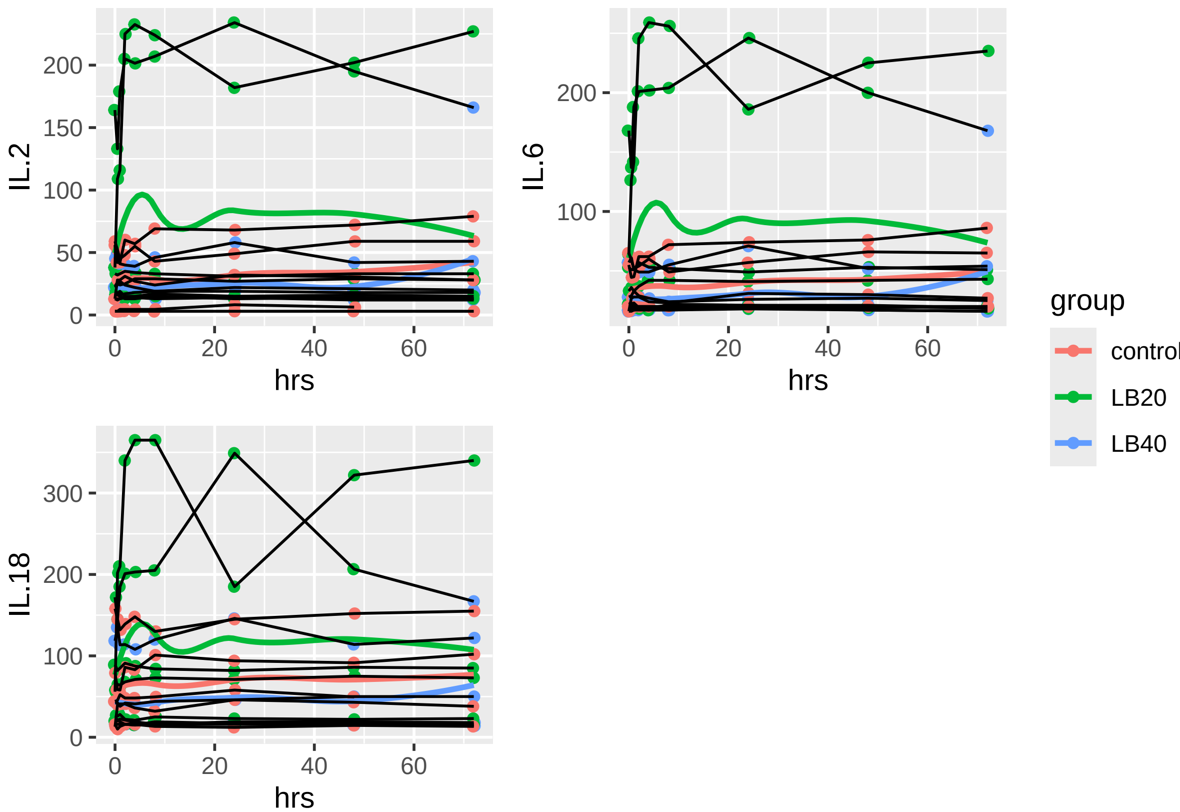
IL.8
300
200
100
0
20
40
60
hrs
group
control
LB20
LB40

There were also differences in a few cytokines at 72 hrs that I believe were unrelated to the study but were in fact related to the analysis - spurious data

These 2 plots look at the aggregate data and individual results




```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



Code used to evaluate and graph the data

```
knitr::opts_chunk$set(echo = FALSE, warning = FALSE)
library(lme4)
library(lmerTest)
library(tidyverse)
library(magrittr)
library(emmeans)
library(kableExtra)
library(ggrepel)
library(outliers)
library(dplyr)

summarySE <- function(data=NULL, measurevar, groupvars=NULL, na.rm=FALSE,
                      conf.interval=.95, .drop=TRUE) {
  library(plyr)
  library(dplyr)

  # New version of length which can handle NA's: if na.rm==T, don't count them
  length2 <- function (x, na.rm=FALSE) {
    if (na.rm) sum(!is.na(x))
    else       length(x)
  }

  # This does the summary. For each group's data frame, return a vector with
```

```r
  # N, mean, and sd
  datac <- ddply(data, groupvars, .drop=.drop,
                 .fun = function(xx, col) {
                   c(N    = length2(xx[[col]], na.rm=na.rm),
                     mean = mean   (xx[[col]], na.rm=na.rm),
                     sd   = sd     (xx[[col]], na.rm=na.rm)
                   )
                 },
                 measurevar
  )

  # Rename the "mean" column
  datac <- rename(datac, c("mean" = measurevar))

  datac$se <- datac$sd / sqrt(datac$N)  # Calculate standard error of the mean

  # Confidence interval multiplier for standard error
  # Calculate t-statistic for confidence interval:
  # e.g., if conf.interval is .95, use .975 (above/below), and use df=N-1
  ciMult <- qt(conf.interval/2 + .5, datac$N-1)
  datac$ci <- datac$se * ciMult

  return(datac)
}

data <- read.table('LL_LP_SAA_cytokine_data_0925.csv', sep = ",", header = TRUE, quote = "")
  #change ID and group to factor variables
data %<>% mutate_at(c("ID", "group"), factor)
head(data)
str(data)
plot1 <- ggplot(data, aes(hrs, saa))
plot1 + geom_jitter(aes(color = group)) +
  geom_smooth(aes(color = group), se = FALSE) +
  geom_line(aes(group = ID))
  # add separator for new controls and older data
data$time <- ifelse(data$ID == 'Vesunna'|
                      data$ID == 'Rita'|
                      data$ID == 'Myla',
                                      'new', 'old') %>% factor()
str(data)
  #test plot
labels1 <- data %>% filter(time == 'new' & hrs == 72)
labels2 <- data %>% filter(ID == 'Rita' & hrs == 48)
  # create separately d/t Rita data ending at 48 hrs
p <- ggplot(data, aes(hrs, Eotaxin)) +
  geom_line(aes(group = ID, color = time)) +
  geom_text_repel(data = labels1, aes(label = ID),
  fontface ="plain", color = "black", size = 3) +
  geom_text_repel(data = labels2, aes(label = ID),
                  fontface ="plain", color = "black", size = 3)
# works as expected
p2 <- ggplot(data, aes(hrs, G.CSF)) +
  geom_line(aes(group = ID, color = time)) +
```

```r
  geom_text_repel(data = labels1, aes(label = ID),
                  fontface ="plain", color = "black", size = 3) +
  geom_text_repel(data = labels2, aes(label = ID),
                  fontface ="plain", color = "black", size = 3)
p3 <- ggplot(data, aes(hrs, IL1a)) +
  geom_line(aes(group = ID, color = time)) +
  geom_text_repel(data = labels1, aes(label = ID),
                  fontface ="plain", color = "black", size = 3) +
  geom_text_repel(data = labels2, aes(label = ID),
                  fontface ="plain", color = "black", size = 3)
p4 <- ggplot(data, aes(hrs, IL.5)) +
  geom_line(aes(group = ID, color = time)) +
  geom_text_repel(data = labels1, aes(label = ID),
                  fontface ="plain", color = "black", size = 3) +
  geom_text_repel(data = labels2, aes(label = ID),
                  fontface ="plain", color = "black", size = 3)
p5 <- ggplot(data, aes(hrs, IL.18)) +
  geom_line(aes(group = ID, color = time)) +
  geom_text_repel(data = labels1, aes(label = ID),
                  fontface ="plain", color = "black", size = 3) +
  geom_text_repel(data = labels2, aes(label = ID),
                  fontface ="plain", color = "black", size = 3)
p6 <- ggplot(data, aes(hrs, IL.6)) +
  geom_line(aes(group = ID, color = time)) +
  geom_text_repel(data = labels1, aes(label = ID),
                  fontface ="plain", color = "black", size = 3) +
  geom_text_repel(data = labels2, aes(label = ID),
                  fontface ="plain", color = "black", size = 3)
p7 <- ggplot(data, aes(hrs, IL.17A)) +
  geom_line(aes(group = ID, color = time)) +
  geom_text_repel(data = labels1, aes(label = ID),
                  fontface ="plain", color = "black", size = 3) +
  geom_text_repel(data = labels2, aes(label = ID),
                  fontface ="plain", color = "black", size = 3)
p8 <- ggplot(data, aes(hrs, IL.2)) +
  geom_line(aes(group = ID, color = time)) +
  geom_text_repel(data = labels1, aes(label = ID),
                  fontface ="plain", color = "black", size = 3) +
  geom_text_repel(data = labels2, aes(label = ID),
                  fontface ="plain", color = "black", size = 3)
p9 <- ggplot(data, aes(hrs, IL.4)) +
  geom_line(aes(group = ID, color = time)) +
  geom_text_repel(data = labels1, aes(label = ID),
                  fontface ="plain", color = "black", size = 3) +
  geom_text_repel(data = labels2, aes(label = ID),
                  fontface ="plain", color = "black", size = 3)
p10 <- ggplot(data, aes(hrs, IFNg)) +
  geom_line(aes(group = ID, color = time)) +
  geom_text_repel(data = labels1, aes(label = ID),
                  fontface ="plain", color = "black", size = 3) +
  geom_text_repel(data = labels2, aes(label = ID),
                  fontface ="plain", color = "black", size = 3)
p11 <- ggplot(data, aes(hrs, IL.8)) +
```

```
  geom_line(aes(group = ID, color = time)) +
  geom_text_repel(data = labels1, aes(label = ID),
                  fontface ="plain", color = "black", size = 3) +
  geom_text_repel(data = labels2, aes(label = ID),
                  fontface ="plain", color = "black", size = 3)
p12 <- ggplot(data, aes(hrs, IP.10)) +
  geom_line(aes(group = ID, color = time)) +
  geom_text_repel(data = labels1, aes(label = ID),
                  fontface ="plain", color = "black", size = 3) +
  geom_text_repel(data = labels2, aes(label = ID),
                  fontface ="plain", color = "black", size = 3)
p13 <- ggplot(data, aes(hrs, IL.10)) +
  geom_line(aes(group = ID, color = time)) +
  geom_text_repel(data = labels1, aes(label = ID),
                  fontface ="plain", color = "black", size = 3) +
  geom_text_repel(data = labels2, aes(label = ID),
                  fontface ="plain", color = "black", size = 3)
p14 <- ggplot(data, aes(hrs, TNFa)) +
  geom_line(aes(group = ID, color = time)) +
  geom_text_repel(data = labels1, aes(label = ID),
                  fontface ="plain", color = "black", size = 3) +
  geom_text_repel(data = labels2, aes(label = ID),
                  fontface ="plain", color = "black", size = 3)

plot_list <- list(p, p2, p3, p4, p5, p6, p7, p8, p9, p10, p11, p12, p13, p14)
combined_plot <- patchwork::wrap_plots(plot_list, nrow=4, guides='collect')
print(combined_plot)
# check time zero for initial outliers
saa_0 <- data %>% filter (hrs == 0) %>% .$saa
dixon.test(saa_0, two.sided=F)
  #yes, value of 1716 at time 0 is an outlier in horse AB
  #Check eotaxin, gcsf and IL8
eotaxin_0 <- data %>% filter (hrs == 0) %>% .$Eotaxin
dixon.test(eotaxin_0, two.sided = F)
gcsf_0 <- data %>% filter (hrs == 0) %>% .$G.CSF
dixon.test(gcsf_0, two.sided = F)
IL8_0 <- data %>% filter (hrs == 0) %>% .$IL.8
dixon.test(IL8_0, two.sided = F)
  #all yes, Myla is an outlier
data2 <- data %>% filter(ID != 'AB' & ID != 'Myla')

data_summary <- data2 %>%
  group_by(hrs, group) %>%
  summarise(mean = mean(saa), sd = sd(saa), min = min(saa), max = max(saa))
#knitr::kable(data_summary)
kbl(data_summary, longtable = T, booktabs = T, align = "c") %>%
  column_spec(1, bold=T) %>%
  collapse_rows(columns = 1:2, latex_hline = "major", valign = "middle")

# create model
model_small <- lmer(saa ~ group*hrs + (1|ID), data = data2)
coef(summary(as(model_small, "lmerModLmerTest"), ddf="Kenward-Roger"))
# note the only real difference is over time, no groups are different in this analysis
```

```r
emm_small2 <- emmeans(model_small, ~ group|hrs, at = list(hrs=c(0,0.5, 1, 2, 4, 8, 24, 48, 72)))
#emm_small2
pairs(emm_small2) # there are small differences in 20 vs 40 at 48 & 72 hrs
                  # with 40 being lower than 20
saa_plot_data <- data %>%
  filter(ID != 'AB' & ID != 'Myla') %>%
  summarySE(measurevar = 'saa', groupvars = c('group', 'hrs'))
# set jitter
pd <- position_dodge(3)
# plot saa
saa_plot <- ggplot(data = saa_plot_data, aes(x = hrs, y = saa, group = group)) +
  geom_errorbar(aes(ymin=saa-ci, ymax=saa+ci), width=2, position=pd) +
  geom_line(aes(linetype = group), position=pd) +
  geom_point(aes(shape = group), size = 3, position=pd) +
  annotate(geom = 'text', x = c(48, 72), y = c(2450, 2350), label = '*', size = 10)
print(saa_plot)
model_eotaxin <- lmer(Eotaxin ~ group*hrs + (1|ID), data = data2)
coef(summary(as(model_eotaxin, "lmerModLmerTest"), ddf="Kenward-Roger")) # Nothing there
emm_eotaxin <- emmeans(model_eotaxin, ~ group|hrs, at = list(hrs=c(0,0.5, 1, 2, 4, 8, 24, 48, 72)))
pairs(emm_eotaxin)
contrast(emm_eotaxin, "trt.vs.ctrl")
eotaxin_plot_data <- data %>%
  filter(ID != 'AB' & ID != 'Myla') %>%
  summarySE(measurevar = 'Eotaxin', groupvars = c('group', 'hrs'))
# set jitter
pd <- position_dodge(3)
# plot
eotaxin_plot <- ggplot(data = eotaxin_plot_data, aes(x = hrs, y = Eotaxin, group = group)) +
  geom_errorbar(aes(ymin=Eotaxin-ci, ymax=Eotaxin+ci), width=2, position=pd) +
  geom_line(aes(linetype = group), position=pd) +
  geom_point(aes(shape = group), size = 3, position=pd)
  #annotate(geom = 'text', x = c(48, 72), y = c(2450, 2350), label = '*', size = 10)
print(eotaxin_plot)
model_gcsf <- lmer(G.CSF ~ group*hrs + (1|ID), data = data2)
coef(summary(as(model_gcsf, "lmerModLmerTest"), ddf="Kenward-Roger")) # Nothing there
emm_gcsf <- emmeans(model_gcsf, ~ group|hrs, at = list(hrs=c(0,0.5, 1, 2, 4, 8, 24, 48, 72)))
pairs(emm_gcsf)
contrast(emm_gcsf, "trt.vs.ctrl")
gcsf_plot_data <- data %>%
  filter(ID != 'AB' & ID != 'Myla') %>%
  summarySE(measurevar = 'G.CSF', groupvars = c('group', 'hrs'))
# set jitter
pd <- position_dodge(3)
# plot
gcsf_plot <- ggplot(data = gcsf_plot_data, aes(x = hrs, y = G.CSF, group = group)) +
  geom_errorbar(aes(ymin=G.CSF-ci, ymax=G.CSF+ci), width=2, position=pd) +
  geom_line(aes(linetype = group), position=pd) +
  geom_point(aes(shape = group), size = 3, position=pd)
  #annotate(geom = 'text', x = c(48, 72), y = c(2450, 2350), label = '*', size = 10)
print(gcsf_plot)
model_IL.8 <- lmer(IL.8 ~ group*hrs + (1|ID), data = data2)
coef(summary(as(model_IL.8, "lmerModLmerTest"), ddf="Kenward-Roger")) # 20 & 40 diff early on
emm_IL.8 <- emmeans(model_IL.8, ~ group|hrs, at = list(hrs=c(0,0.5, 1, 2, 4, 8, 24, 48, 72)))
```

```
pairs(emm_IL.8)
contrast(emm_IL.8, "trt.vs.ctrl")

IL8_plot_data <- data %>%
  filter(ID != 'AB' & ID != 'Myla') %>%
  summarySE(measurevar = 'IL.8', groupvars = c('group', 'hrs'))
# set jitter
pd <- position_dodge(3)
# plot
IL8_plot <- ggplot(data = IL8_plot_data, aes(x = hrs, y = IL.8, group = group)) +
  geom_errorbar(aes(ymin=IL.8-se, ymax=IL.8+se), width=2, position=pd) +
  geom_line(aes(linetype = group), position=pd) +
  geom_point(aes(shape = group), size = 3, position=pd) +
  annotate(geom = 'text', x = c(0, 0.5, 1, 2, 4, 8, 24), y = c(275, 300, 325, 350, 350, 350, 350), label
print(IL8_plot)
IL18_plot_data <- data %>%
  filter(ID != 'AB' & ID != 'Myla') %>%
  summarySE(measurevar = 'IL.18', groupvars = c('group', 'hrs'))
# set jitter
pd <- position_dodge(3)
# plot
IL18_plot <- ggplot(data = IL18_plot_data, aes(x = hrs, y = IL.18, group = group)) +
  geom_errorbar(aes(ymin=IL.18-ci, ymax=IL.18+ci), width=2, position=pd) +
  geom_line(aes(linetype = group), position=pd) +
  geom_point(aes(shape = group), size = 3, position=pd) +
  annotate(geom = 'text', x = 72, y = 280, label = '*', size = 10)
#print(IL18_plot)
IL18_raw <- ggplot(data2, aes(hrs, IL.18))
IL18_raw_plot <- IL18_raw + geom_jitter(aes(color=group)) +
  geom_smooth(aes(color=group), se=FALSE) +
  geom_line(aes(group=ID))

IL6_plot_data <- data %>%
  filter(ID != 'AB' & ID != 'Myla') %>%
  summarySE(measurevar = 'IL.6', groupvars = c('group', 'hrs'))
# set jitter
pd <- position_dodge(3)
# plot
IL6_plot <- ggplot(data = IL6_plot_data, aes(x = hrs, y = IL.6, group = group)) +
  geom_errorbar(aes(ymin=IL.6-se, ymax=IL.6+se), width=2, position=pd) +
  geom_line(aes(linetype = group), position=pd) +
  geom_point(aes(shape = group), size = 3, position=pd) +
  annotate(geom = 'text', x = 72, y = 125, label = '*', size = 10)
#print(IL6_plot)
IL6_raw <- ggplot(data2, aes(hrs, IL.6))
IL6_raw_plot <- IL6_raw + geom_jitter(aes(color=group)) +
  geom_smooth(aes(color=group), se=FALSE) +
  geom_line(aes(group=ID))

IL2_plot_data <- data %>%
  filter(ID != 'AB' & ID != 'Myla') %>%
  summarySE(measurevar = 'IL.2', groupvars = c('group', 'hrs'))
# set jitter
```

```r
pd <- position_dodge(3)
# plot
IL2_plot <- ggplot(data = IL2_plot_data, aes(x = hrs, y = IL.2, group = group)) +
  geom_errorbar(aes(ymin=IL.2-se, ymax=IL.2+se), width=2, position=pd) +
  geom_line(aes(linetype = group), position=pd) +
  geom_point(aes(shape = group), size = 3, position=pd) +
  annotate(geom = 'text', x = 72, y = 110, label = '*', size = 10)
#print(IL2_plot)
IL2_raw <- ggplot(data2, aes(hrs, IL.2))
IL2_raw_plot <- IL2_raw + geom_jitter(aes(color=group)) +
  geom_smooth(aes(color=group), se=FALSE) +
  geom_line(aes(group=ID))

IL.2.6.18_plot <- patchwork::wrap_plots(IL2_plot, IL6_plot, IL18_plot,
                                        nrow=2, guides='collect')
print(IL.2.6.18_plot)

IL.2.6.18_raw_plot <- patchwork::wrap_plots(IL2_raw_plot, IL6_raw_plot, IL18_raw_plot,
                                            nrow=2, guides='collect')
print(IL.2.6.18_raw_plot)
```